PRACTICAL EXAMINATION AND ORAL (VIVA) QUESTION BANK

Simple English • Subject-wise • Based on your uploaded practical files

Subjects covered

1. Design and Analysis of Algorithms (DAA)
2. Machine Learning (ML)
3. Blockchain Technology (BCT)

Includes practical explanations, important viva questions, short answers, algorithms, key terms and quick revision points.

Prepared in easy-to-understand English for practical and oral examination preparation.

1. DESIGN AND ANALYSIS OF ALGORITHMS (DAA)

This section is based on the four uploaded DAA practicals: Fibonacci (recursive and non-recursive), Huffman Encoding, 0-1 Knapsack using Dynamic Programming, and Fractional Knapsack using a Greedy method.

Practical 1: Fibonacci Series — Recursive and Non-Recursive

Q. What is the Fibonacci series?
It is a series in which each number is the sum of the previous two numbers. It starts with 0 and 1.

Q. Write the first few Fibonacci numbers.
0, 1, 1, 2, 3, 5, 8, 13, 21, 34.

Q. What is recursion?
Recursion is a method in which a function calls itself to solve a smaller part of a problem.

Q. What is the base condition in the recursive Fibonacci program?
If n is 0 or 1, the function returns n. This stops further recursive calls.

Q. What is a non-recursive or iterative method?
It uses a loop to calculate the answer instead of calling the same function again.

Q. Which variables are used in the iterative program?
The variables a and b store the previous two numbers, and c stores their sum.

Q. What is the time complexity of recursive Fibonacci?
The simple recursive method takes exponential time, commonly written as O(2^n), because many calls are repeated.

Q. What is the space complexity of recursive Fibonacci?
O(n) due to the function call stack.

Q. What is the time complexity of iterative Fibonacci?
O(n), because the loop runs n times.

Q. What is the extra space complexity of iterative Fibonacci?
O(1), because it uses only a fixed number of variables.

Q. Which method is faster for this practical and why?
The iterative method is faster because it avoids repeated recursive calls.

Q. What happens if the input number of terms is 5?
The output is 0 1 1 2 3.

Practical 2: Huffman Encoding using Greedy Strategy

Q. What is Huffman Encoding?
Huffman Encoding is a lossless data compression technique. It gives shorter binary codes to more frequent characters.

Q. What is lossless compression?
It reduces the size of data without losing the original information.

Q. What is a greedy algorithm?
A greedy algorithm makes the best available choice at each step.

Q. Why is Huffman Encoding called a greedy algorithm?
At every step, it selects the two nodes with the lowest frequencies.

Q. Which data structure is used in the practical?
A PriorityQueue is used to select the node with the smallest frequency.

Q. What is a node in the Huffman tree?
A node stores a character, its frequency, and links to its left and right child nodes.

Q. How is a new internal node created?
The two smallest-frequency nodes are removed and combined. Their frequencies are added to create a new node.

Q. When does the combining process stop?
It stops when only one node remains in the priority queue. That node is the root of the Huffman tree.

Q. How are binary codes generated?
The program adds 0 when moving left and 1 when moving right. A character's code is the path from root to its leaf.

Q. What is a leaf node?
A node with no left or right child. In this program, each leaf represents a character.

Q. Why do frequent characters get shorter codes?
They are placed closer to the root, so their paths contain fewer bits.

Q. What are the sample frequencies in the uploaded program?
A: 5, B: 9, C: 12, D: 13, E: 16 and F: 45.

Q. Give the sample Huffman codes shown in the practical.
F: 0, C: 100, D: 101, A: 1100, B: 1101 and E: 111.

Q. What is the main benefit of Huffman Encoding?
It can reduce the total number of bits needed to store or transmit data.

Practical 3: 0-1 Knapsack using Dynamic Programming

Q. What is the 0-1 Knapsack problem?
We are given items with values and weights. We must choose items to get the maximum value without exceeding the bag capacity.

Q. Why is it called 0-1 Knapsack?
Each item is either selected completely (1) or not selected (0). We cannot take a fraction of an item.

Q. Which technique is used in the uploaded program?
Dynamic Programming.

Q. What is Dynamic Programming?
It solves a problem by dividing it into smaller overlapping subproblems and storing their results.

Q. What does dp[i][w] represent?
It represents the maximum value possible using the first i items and a capacity of w.

Q. What are the two choices for each item?
Take the item if it fits, or do not take the item.

Q. What is the recurrence relation used?
If the item fits: dp[i][w] = max(value[i-1] + dp[i-1][w-weight[i-1]], dp[i-1][w]). Otherwise, dp[i][w] = dp[i-1][w].

Q. What is the base condition?
The first row and first column are zero. With zero items or zero capacity, the maximum value is zero.

Q. How does the program find selected items?
It checks whether dp[i][w] differs from dp[i-1][w]. If it differs, item i was selected and its weight is subtracted.

Q. What is the sample capacity in the uploaded output?
60.

Q. What is the maximum value in the sample output?
270, by selecting item 1 (value 120, weight 20) and item 2 (value 150, weight 40).

Q. What is the time complexity of this Dynamic Programming solution?
O(n × capacity), where n is the number of items.

Q. What is the space complexity?
O(n × capacity), because a two-dimensional DP table is used.

Practical 4: Fractional Knapsack using Greedy Method

Q. What is the Fractional Knapsack problem?
It is a problem where items have values and weights, and we can take a whole item or a fraction of it.

Q. Which method is used?
The Greedy method.

Q. How are items selected?
Calculate value divided by weight for each item, then select items in decreasing order of this ratio.

Q. What is the value/weight ratio?
It tells us the value gained for one unit of weight.

Q. Why do we sort items by ratio?
To take the item that gives the highest value per unit of weight first.

Q. Can we take part of an item?
Yes. Fractional Knapsack allows taking a fraction of an item.

Q. What happens when the next item does not fit completely?
The program takes only the fraction that fits in the remaining capacity.

Q. What is the capacity in the uploaded Java program?
60.

Q. What is the sample maximum value?
245.00.

Q. Which items are taken in the sample output?
Item 1 and Item 2 are taken completely, and 0.17 fraction of Item 3 is taken.

Q. What is the difference between 0-1 and Fractional Knapsack?
In 0-1 Knapsack, an item is either fully taken or rejected. In Fractional Knapsack, part of an item can also be taken.

Q. What is the time complexity of the greedy Fractional Knapsack method?
O(n log n), mainly because of sorting the items.

Important DAA General Viva Questions

Q. What is an algorithm?
An algorithm is a finite set of clear steps used to solve a problem.

Q. What is time complexity?
It describes how the running time of an algorithm grows as the input size increases.

Q. What is space complexity?
It describes how much extra memory an algorithm needs.

Q. What is Big-O notation?
Big-O describes an upper bound on the growth of an algorithm's time or space requirement.

Q. What is the difference between Greedy and Dynamic Programming?
Greedy makes a locally best choice at each step. Dynamic Programming stores solutions to smaller subproblems and combines them.

Q. What is sorting?
Sorting means arranging data in a particular order, such as ascending or descending.

Q. What is a Priority Queue?
It is a data structure where elements are removed according to priority, such as smallest frequency first.

Q. What is a tree?
A tree is a non-linear data structure made of nodes connected by edges.

2. MACHINE LEARNING (ML)

This section is based on the four uploaded ML practicals: diabetes prediction using KNN, bank customer churn prediction using ANN, spam email classification using KNN and SVM, and Uber fare prediction using Linear Regression and Random Forest.

Practical 1: Diabetes Prediction using KNN

Q. What is Machine Learning?
Machine Learning is a part of Artificial Intelligence that allows computers to learn patterns from data and make predictions.

Q. What is the aim of this practical?
To predict the diabetes outcome using the K-Nearest Neighbors classifier.

Q. What is KNN?
K-Nearest Neighbors is a supervised learning algorithm. It predicts a class by looking at the nearest data points.

Q. What does K mean in KNN?
K is the number of nearest neighbors considered for prediction.

Q. What K value is used in the main model?
K = 5.

Q. What dataset is used?
The program loads diabetes.csv.

Q. What is a feature?
A feature is an input column used by the model to make a prediction.

Q. What is the target column?
Outcome is the target column in this practical.

Q. What is data cleaning?
Data cleaning means finding and handling missing, incorrect or unsuitable values.

Q. Why are some zero values replaced with the mean?
In the listed columns, zero is treated as a missing or invalid value and is replaced with the column mean.

Q. Which columns have zero values replaced?
Glucose, BloodPressure, SkinThickness, Insulin and BMI.

Q. What is train_test_split?
It divides the dataset into training data and testing data.

Q. What test size is used?
20 percent of the data is used for testing.

Q. What is a boxplot used for?
A boxplot helps visualize the spread of data and possible outliers.

Q. What is IQR?
IQR means Interquartile Range. It is Q3 minus Q1.

Q. How are outliers identified in this program?
Values below Q1 - 1.5 × IQR or above Q3 + 1.5 × IQR are counted as outliers.

Q. What is accuracy?
Accuracy is the number of correct predictions divided by the total number of predictions.

Q. What is precision?
Precision tells us how many predicted positive cases are actually positive.

Q. What is recall?
Recall tells us how many actual positive cases were correctly found.

Q. What is F1 score?
F1 score is the harmonic mean of precision and recall.

Q. What is a confusion matrix?
It is a table that compares actual classes with predicted classes.

Q. Which K values are compared?
K = 3, 5 and 7.

Practical 2: Bank Customer Churn Prediction using ANN

Q. What is customer churn?
Customer churn means a customer leaves or stops using a company's service.

Q. What is the aim of this practical?
To predict whether a bank customer will leave the bank.

Q. What is ANN?
An Artificial Neural Network is a machine learning model inspired by the way connected neurons work.

Q. Which library is used to build the ANN?
TensorFlow Keras.

Q. Which dataset is loaded?
Bank.csv.

Q. What is the target column?
Exited.

Q. Why are RowNumber, CustomerId and Surname removed?
They are unnecessary identification columns for this prediction model.

Q. What is categorical data?
It is data represented by groups or labels, such as Geography and Gender.

Q. What is one-hot encoding?
It converts categorical values into numeric columns so a model can use them.

Q. Which function is used for encoding?
pandas.get_dummies().

Q. What does drop_first=True do?
It drops the first category column to avoid redundant dummy columns.

Q. What is feature scaling?
It changes feature values to a common scale.

Q. Which scaler is used?
StandardScaler.

Q. Why is the dataset split?
Training data is used to teach the model, while testing data is used to evaluate it.

Q. What test size is used?
20 percent.

Q. What is a Sequential model?
It is a Keras model in which layers are added one after another.

Q. What is a Dense layer?
A Dense layer connects each neuron to all outputs from the previous layer.

Q. What is the ReLU activation function?
ReLU returns zero for negative inputs and the input value for positive inputs.

Q. What is the Sigmoid activation function?
It gives an output between 0 and 1, useful for binary classification.

Q. What is Dropout?
Dropout randomly disables some neurons during training to help reduce overfitting.

Q. What is overfitting?
Overfitting happens when a model learns the training data too closely and performs poorly on new data.

Q. What is the Adam optimizer?
Adam is an optimization method used to update model weights during training.

Q. What is binary cross-entropy?
It is a loss function commonly used for binary classification.

Q. What is an epoch?
One epoch means one complete pass through the training data.

Q. What is batch size?
It is the number of training samples processed before the model updates its weights.

Q. What is EarlyStopping?
It stops training when the monitored validation result does not improve for a set number of epochs.

Q. What threshold is used for prediction?
0.5. A probability of 0.5 or more is classified as 1.

Q. What does the Exited value mean?
0 means the customer stayed and 1 means the customer exited, as shown by the practical's class labels.

Practical 3: Spam Email Classification using KNN and SVM

Q. What is spam email?
Spam is an unwanted or unsolicited email, often sent in bulk.

Q. What is the aim of this practical?
To classify emails as Not Spam or Spam using KNN and SVM.

Q. Which dataset is used?
emails.csv.

Q. Which columns are removed from the input features?
Email No. and Prediction.

Q. What is the target column?
Prediction.

Q. How are missing feature values handled?
They are replaced with zero.

Q. What are the two classes?
0 represents Not Spam and 1 represents Spam in the practical labels.

Q. What is KNN?
KNN classifies a new sample using the classes of its nearest data points.

Q. What K value is used?
K = 5.

Q. What is SVM?
Support Vector Machine is a supervised learning algorithm that finds a decision boundary between classes.

Q. Which SVM kernel is used?
The linear kernel.

Q. What is a kernel in SVM?
A kernel helps SVM create a decision boundary. The practical uses a linear kernel.

Q. What is stratify=y?
It keeps the class proportions similar in the training and testing sets.

Q. What is classification?
Classification is the task of predicting a category or class.

Q. Why are KNN and SVM compared?
To compare their performance using accuracy, precision, recall and F1 score.

Q. What is a classification report?
It displays precision, recall, F1 score and support for each class.

Q. What is support in a classification report?
Support is the number of actual samples belonging to a class.

Q. What does zero_division=0 mean?
It returns zero for a metric when its calculation has a zero denominator, instead of raising a warning.

Q. How does the program choose the better model?
It compares the F1 scores of KNN and SVM.

Practical 4: Uber Fare Prediction using Regression

Q. What is regression?
Regression is a machine learning task used to predict a continuous numeric value.

Q. What is the aim of this practical?
To predict Uber fare amount using trip details.

Q. Which dataset is used?
uber.csv.

Q. What is the target variable?
fare_amount.

Q. Which models are compared?
Linear Regression and Random Forest Regressor.

Q. What is Linear Regression?
It predicts a numeric value by learning a linear relationship between input features and the target.

Q. What is Random Forest Regressor?
It combines many decision trees to predict a numeric value.

Q. Why are missing rows removed?
Rows with missing values are removed so the model can work with complete records.

Q. Why are latitude and longitude checked?
To remove coordinates outside valid geographic ranges.

Q. Why are rows with fare_amount less than or equal to zero removed?
A fare should be positive for this prediction task.

Q. What date-time features are extracted?
Hour, day, month, year and day_of_week.

Q. What is feature engineering?
It means creating useful input features from existing data.

Q. What is the Haversine formula used for?
It calculates the distance between two points on the Earth using latitude and longitude.

Q. What is distance_km?
It is the calculated trip distance in kilometres.

Q. What is an outlier?
An outlier is a value that is unusually far from most other values.

Q. How does the practical identify fare and distance outliers?
It uses the IQR method and removes values outside the lower and upper bounds.

Q. What is correlation?
Correlation measures how two numeric variables move in relation to each other.

Q. What does StandardScaler do?
It standardizes features by removing the mean and scaling to unit variance.

Q. What is R-squared (R2)?
It measures how well the model explains the variation in the target. A higher value is generally better.

Q. What is MAE?
Mean Absolute Error is the average absolute difference between actual and predicted values. Lower is better.

Q. What is RMSE?
Root Mean Squared Error is the square root of the average squared prediction error. Lower is better.

Q. Why compare Linear Regression and Random Forest?
To find which model gives better prediction results on the test data.

Q. What does n_estimators=100 mean?
The Random Forest uses 100 decision trees.

Important ML General Viva Questions

Q. What is supervised learning?
Learning from labelled examples where the correct output is provided.

Q. What is unsupervised learning?
Learning patterns from data without given target labels.

Q. What is classification?
Predicting a class or category.

Q. What is regression?
Predicting a continuous numeric value.

Q. What is a dataset?
A collection of data used for analysis or model training.

Q. What is a training dataset?
The part of data used to train a model.

Q. What is a testing dataset?
The part of data used to check how well a trained model performs on unseen data.

Q. What is a model?
A learned mathematical or computational representation used to make predictions.

Q. What is prediction?
The output produced by a trained model for new input data.

Q. What is a metric?
A measure used to evaluate model performance.

3. BLOCKCHAIN TECHNOLOGY (BCT)

This section is based on the uploaded Solidity Bank smart contract. It includes deposit, withdrawal and balance checking functions.

Practical: Bank Smart Contract using Solidity

Q. What is Blockchain?
Blockchain is a digital record system in which data is stored in linked blocks across a network.

Q. What is a smart contract?
A smart contract is a program stored on a blockchain that runs when its functions are called.

Q. What is Solidity?
Solidity is a programming language used to write smart contracts, especially for Ethereum-compatible blockchains.

Q. What is the name of the contract in the practical?
Bank.

Q. Which Solidity version is specified?
The program specifies pragma solidity ^0.8.0.

Q. What does pragma solidity mean?
It tells the compiler which Solidity compiler version range the program is intended to use.

Q. What is a mapping in Solidity?
A mapping stores key-value pairs. In this program, an address is linked to a uint256 balance.

Q. What does mapping(address => uint256) public balances mean?
It stores a numeric balance for each blockchain address and makes the mapping publicly readable.

Q. What is uint256?
It is an unsigned integer type that can store a large non-negative whole number.

Q. What is msg.sender?
It is the address that called the current smart contract function.

Q. What does depositMoney() do?
It adds the given amount to the balance of the caller.

Q. Why does depositMoney() use require(amount > 0)?
It prevents a zero or negative-style deposit. The amount must be greater than zero.

Q. What does withdrawMoney() do?
It subtracts the requested amount from the caller's balance.

Q. Why is the balance checked before withdrawal?
To make sure the caller has enough balance and cannot withdraw more than the available amount.

Q. What happens if the balance is insufficient?
The require statement fails with the message 'Insufficient balance'.

Q. What does showBalance() do?
It returns the balance of the caller.

Q. Why is showBalance() marked view?
Because it only reads contract data and does not modify the blockchain state.

Q. What does public mean in a function declaration?
It allows the function to be called externally and internally, subject to Solidity rules.

Q. What is a function?
A function is a named block of code that performs a particular task.

Q. What is require()?
It checks a condition. If the condition is false, execution stops and the transaction is reverted.

Q. What is a transaction?
A transaction is an operation submitted to the blockchain, such as calling a function that changes state.

Q. Which functions change the balance?
depositMoney() and withdrawMoney().

Q. Which function only reads the balance?
showBalance().

Q. Does the contract store one common balance for everyone?
No. The mapping stores a separate balance for each address.

Q. What is the purpose of the SPDX line?
It declares the source-code licence. The practical uses MIT.

Q. What is the difference between read and write functions?
A read function checks data without changing state. A write function changes the contract state.

Quick Practical Explanation — How to Explain Your Programs in the Exam

- DAA Fibonacci: The program accepts the number of terms and prints the Fibonacci series. The recursive version calls itself; the iterative version uses a loop.
- DAA Huffman: The program stores characters with frequencies, repeatedly combines the two lowest frequencies, builds a tree and prints binary codes.
- DAA 0-1 Knapsack: The program accepts item values, weights and capacity, fills a DP table and prints the maximum value and selected items.
- DAA Fractional Knapsack: The program calculates value/weight ratios, sorts items in descending order and takes whole items or a fraction until the bag is full.
- ML Diabetes KNN: The program loads diabetes data, handles selected zero values, splits the data, trains KNN and evaluates predictions.
- ML Bank ANN: The program prepares bank customer data, encodes categories, scales features, trains an ANN and evaluates customer churn predictions.
- ML Spam: The program prepares email data, trains KNN and linear SVM classifiers, and compares their evaluation scores.
- ML Uber Fare: The program cleans trip data, calculates distance and time features, trains two regression models and compares R2, RMSE and MAE.
- BCT Bank: The contract keeps balances by wallet address and provides functions to deposit money, withdraw money and show the caller's balance.

Last-Minute Viva Revision

- Fibonacci: Recursive O(2^n) time; iterative O(n) time.
- Huffman: Greedy strategy + Priority Queue + binary tree.
- 0-1 Knapsack: Dynamic Programming; no fractions allowed.
- Fractional Knapsack: Greedy strategy; fractions are allowed; sort by value/weight.
- KNN: Uses nearest neighbours; K is the number of neighbours.
- ANN: Neural network with layers; sigmoid output for binary prediction in this practical.
- SVM: Finds a decision boundary; linear kernel is used in the spam practical.
- Regression: Predicts numeric values; R2 higher is generally better, MAE/RMSE lower is better.

- Blockchain: Distributed digital record; Solidity is used for smart contracts.
- Smart contract: depositMoney(), withdrawMoney(), showBalance(); msg.sender identifies the caller.

Exam Preparation Notes

How to answer in the practical oral examination

- First explain the aim of the practical in one sentence.
- Name the dataset, programming language or main library used.
- Explain the main steps in the correct order.
- Be ready to explain important variables, functions and parameters.
- Explain the output and what it means.
- If asked about performance, state the relevant time complexity or evaluation metrics.
- Do not memorize unknown output values. The ML model scores can change depending on the dataset and environment.

Source note: The questions and answers are prepared from the nine practical files uploaded in this chat. Short background definitions are included to help explain the practicals in simple English. Where the uploaded files do not show a fixed ML score, this guide does not invent one.